

ASTITVA PATHAK

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EDUCATION

Rajiv Gandhi Institute of Petroleum Technology (RGPT)

Bachelor of Technology in Information Technology – CGPA: 7.5/10

Jais, India

2022 – 2026

Pragati Vidhya Peeth

Intermediate – 87.6%

Matriculation – 90.8%

EXPERIENCE

Software Engineer I – Rayvector Technologies

Full-Time | May 2026 – Present

- Built streaming, recording, and real-time inference pipeline; cut inference time by 50%.
- Wrote end-to-end Nvidia hardware tool in C++ using Deepstream and GStreamer with a queue-based alarm system; rebuilt prior architecture in half the original time.
- Tuned memory management and resource allocation for long-running processes on Jetson platforms, maintaining 99.9% uptime in production.

Software Engineer Intern – Rayvector Technologies

Jan 2026 – Apr 2026

- Trained and deployed Computer Vision models for semiconductor manufacturing; improved anomaly detection accuracy by 50% across production lines.
- Built hardware-software solution for real-time silicon wafer thickness monitoring during etching and deposition, tightening process control precision.
- Integrated Raspberry Pi, Jetson, and NVIDIA edge controllers into production-grade manufacturing pipelines.

DevOps & AI Engineer – Audio

Part-Time | Aug 2025 – Jan 2026

- Set up Azure CI/CD pipeline with Jenkins; cut deployment time by 40% and increased release cadence.
- Built agentic AI pipeline to transcribe, summarize, and generate sentiment-aligned audio for videos; handled 500+ videos/day at 95% accuracy.
- Ran containerized media services on Docker and Kubernetes, sustaining peaks of 10,000+ concurrent requests.

Full Stack Developer – Audio

Part-Time | Jun 2025 – Jul 2025

- Built full-stack AI music generator with Next.js, TypeScript, and MongoDB; grew user engagement 10x post-launch.
- Developed timeline-based audio/video editor supporting 1,000+ concurrent sessions with real-time collaboration.
- Used FFmpeg for subtitle overlays, image compositing, and 4K rendering; reduced export time by 65%.
- Added Framer Motion micro-interactions; achieved 95% user satisfaction and cut bounce rate by 40%.

Frontend Developer – Tutpoint.ai

Jan 2024 – Apr 2024

- Built responsive React.js frontend with GSAP ScrollTrigger animations; increased engagement 5x and session duration by 120%.
- Wired real-time problem-solving interface over WebSocket, handling 200+ concurrent users.
- Cut page load time by 60% via code splitting, lazy loading, and responsive design; improved Core Web Vitals.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, Bash, C

Web Development: React.js, Next.js, Node.js, Express.js, MongoDB, PostgreSQL, REST APIs

DevOps & Cloud: Docker, Kubernetes, Jenkins, Azure CI/CD, Prometheus, Grafana, AWS

AI/ML: Computer Vision, OpenCV, TensorFlow, PyTorch, CNN, RNN, Librosa

Security: OWASP ZAP, Burp Suite, Metasploit, Nmap, Wireshark, Penetration Testing

Tools & Technologies: FFmpeg, Janus WebRTC, GStreamer, Git, Linux, Embedded Systems

ACHIEVEMENTS & LEADERSHIP

- Tech Lead, E-Cell RGPT** – Led development of 5+ technical projects impacting 1,000+ students; coordinated hackathons and workshops.
- Full Tuition Scholarship** – Merit-based scholarship for academic excellence throughout undergraduate studies.
- JEE Advanced 2022** – Top 2% among 250,000+ candidates nationwide.
- Community Mentorship** – Mentored 200+ students in rural Uttar Pradesh in technology and career guidance.